

Financial Turbulence Platform: A Risk-Managed, Alpha-Tilted Multi-Asset Strategy

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Abstract

We build a multi-asset allocation system that uses a statistical turbulence signal to scale portfolio risk, a minimum-variance/vol-targeting base for stability, a leak-free exceedance classifier as a leading de-risk trigger, and a time-series momentum tilt for return. Every component is validated under a leakage-safe walk-forward backtest, net of transaction costs, on ETF data spanning 2010–2026. A naively turbulence-tilted Black–Litterman strategy that is *dominated* by equal-weight (Sharpe 0.27) is rebuilt into a strategy with Sharpe 0.82, Sortino 1.05, and a maximum drawdown of -8.8% on an expanded 14-asset universe. The work emphasises honest evaluation: a deep LSTM-CNN forecaster is shown to be statistically indistinguishable from predicting the mean and is discarded in favour of a logistic regression that genuinely beats persistence.

1 Motivation

Diversification fails precisely when it is most needed: asset correlations spike during crises, so a portfolio that looks diversified in calm markets becomes concentrated in a single “risk-on/off” factor during a drawdown. The *turbulence index* of Kritzman and Li (2010) measures this directly, and the goal of this project is to turn it into a disciplined risk-scaling and allocation system—and to do so with the methodological rigour (leakage control, honest baselines, robustness testing) that separates a credible quant strategy from an over-fit backtest.

2 Data and Universe

The core universe is ten liquid ETFs spanning equities (SPY, EFA, EEM), real estate (VNQ), rates and credit (AGG, LQD, HYG, TLT), and commodities (GLD, DBC), 2010–2026, daily. An *expanded* 14-asset universe adds four distinct, low-correlation sleeves—T-bills (BIL), TIPS (TIP), oil (USO), and EM bonds (EMB).

Effective breadth. Eigenanalysis of the return correlation matrix shows the first principal component (the common risk factor) explains 43.9% of variance and the first three explain 82%. The *effective number of independent bets* (entropy of normalised eigenvalues) is only 4.89 out of 10, and it *collapses to 3.80 during the COVID crash*—a quantitative statement of why crisis diversification evaporates. Adding the four distinct sleeves raises effective breadth to 6.54, confirming they are genuinely diversifying rather than redundant.

3 Methodology

3.1 Turbulence index

For a return vector x_t with rolling mean μ and covariance Σ (estimated on a trailing window, strictly in-sample), turbulence is the squared Mahalanobis distance

$$d_t^2 = (x_t - \mu)^\top \Sigma^{-1} (x_t - \mu), \quad (1)$$

equivalently a one-sample Hotelling T^2 statistic. High d_t^2 means today’s cross-asset move is statistically improbable under the recent covariance regime.

3.2 Allocation base

Expected returns are the noisiest input to mean–variance optimisation, and inverting a raw sample covariance amplifies estimation error (“error maximisation”). We therefore (i) shrink the covariance with the Ledoit–Wolf estimator and (ii) use a *risk-based* base—global minimum-variance with a per-asset cap (risk parity and inverse-vol are also implemented)—which needs no return forecast.

3.3 Volatility targeting and the turbulence overlay

The base book is scaled to a fixed annualised volatility target, then a turbulence *exposure overlay* scales gross exposure toward a cash sleeve (earning the risk-free rate) when the trailing percentile of turbulence is elevated. This is the Kritzman–Li signal used as intended—to scale *risk*, not to rotate cross-sectionally into bonds.

3.4 Forecasting: exceedance, not level

Forecasting the turbulence *level* is futile: its autocorrelation decays from 0.48 (1d) to 0.16 (30d) to -0.02 (90d), so a level regressor merely re-derives persistence. A deep LSTM-CNN, evaluated leak-free, does not beat a persistence baseline (7-day level rank-IC 0.146 vs persistence 0.175) and is statistically indistinguishable from predicting the training mean. We reframe the task as *exceedance classification*—will turbulence cross its trailing 90th percentile within k days?—and find a regularised logistic regression beats persistence on out-of-sample AUC (Table ??); a gradient-boosted model over-fits. The deep model is discarded.

Table 1: Out-of-sample exceedance AUC vs a persistence baseline.

Horizon	Persistence	Logistic regression	Gradient boosting
$k = 5$	0.701	0.759	0.60
$k = 10$	0.685	0.727	0.50
$k = 21$	0.668	0.693	0.59

3.5 Time-series momentum tilt

Finally we add a return signal: time-series momentum (Moskowitz, Ooi & Pedersen 2012). Each asset’s risk-adjusted 12-minus-1-month trend tilts the risk-based base (continuous tilt or a long-flat sign filter). Beyond adding alpha, momentum is a *principled* re-entry mechanism—it re-risks into recoveries and shrinks persistent downtrends.

3.6 Backtest protocol

All results use a walk-forward backtest with monthly rebalancing, 10 bps one-way costs, and a strict no-lookahead guard (weights at rebalance i use only data through $i-1$); the guard is unit tested. Performance is summarised by annualised Sharpe and Sortino, annualised volatility, maximum drawdown, and one-way turnover.

4 Results

4.1 Building up the strategy

Table ?? isolates each layer’s contribution on the core universe. The turbulence overlay adds +0.05 Sharpe over naive vol-targeting; the forecast-driven overlay lifts Sortino from 0.82 to 0.92 at lower turnover; momentum and the expanded universe compound to the best result.

Table 2: Incremental results (net of costs, leakage-safe). Top: core 10-asset universe, full period. Bottom: expanded 14-asset universe.

Configuration	Sharpe	Sortino	Ann. vol	Max DD	Turnover
<i>Core universe (10 assets)</i>					
Equal-weight	0.563	0.692	0.092	−0.218	0.03
Min-var + vol-target	0.644	0.767	0.051	−0.170	0.43
+ turbulence overlay	0.696	0.904	0.042	−0.160	1.82
+ momentum (sign)	0.740	0.943	0.042	−0.102	2.62
<i>Expanded universe (14 assets)</i>					
Equal-weight	0.443	0.532	0.086	−0.234	0.03
Strategy (no momentum)	0.714	0.912	0.025	−0.103	1.83
+ momentum (continuous)	0.822	1.048	0.024	−0.088	2.02

4.2 Marginal value of the forecast

Over the common range 2013–2026, swapping the contemporaneous turbulence signal for the trailing percentile of the forecast exceedance probability improves the overlay: Sharpe $0.63 \rightarrow 0.69$, Sortino $0.82 \rightarrow 0.92$, max drawdown $-16\% \rightarrow -15\%$, with *lower* turnover (1.28 vs 1.82)—the gain is better timing, not more trading.

4.3 Robustness

The momentum tilt-strength sweep is smooth and monotonic (Sharpe $0.71 \rightarrow 0.83$ as strength goes $0 \rightarrow 1$ on the expanded universe), the signature of a real effect rather than a fitted spike. By sub-period (Sharpe), the continuous tilt is the more balanced choice: the sign filter helps the 2022 bear strongly but is weaker in 2016–2020, whereas the continuous tilt improves every sub-period without large regressions.

5 Crisis Stress Tests

Table ?? reports the best config (expanded universe + continuous momentum) through six stress episodes. The strategy cushions every crisis—most strikingly COVID (−3.7% drawdown vs SPY −35.7%) and the 2026 Iran-war oil shock (−0.7% vs SPY −7.8%), a supply-driven dislocation it was never tuned for, detected because turbulence keys off cross-asset correlation breakdown.

Table 3: Maximum drawdown within crisis windows (expanded universe + momentum).

Episode	SPY	Equal-weight	Strategy (base)	Strategy (+mom)
2011 Euro crisis	−19.5%	−8.8%	−1.5%	−1.8%
2015–16 China/oil	−13.2%	−10.9%	−2.0%	−2.0%
2018 Q4 selloff	−19.8%	−9.1%	−1.1%	−1.0%
COVID crash	−35.7%	−23.4%	−4.5%	−3.7%
2022 bear (rates)	−26.2%	−16.6%	−9.4%	−8.3%
2026 Iran war oil shock	−7.8%	−3.0%	−0.9%	−0.7%

6 Limitations

We report these candidly, as they are part of the analysis:

- **Low absolute return.** The book is deliberately low-volatility ($\sim 2.4\%$), so annualised return is modest ($\sim 2\%$); the value is risk-adjusted, not headline return.
- **2022 is the hard regime.** When bonds and equities fall together, the minimum-variance base—which leans on bonds—offers less protection (−8.3%).
- **Fast crashes cannot be fully dodged.** A 5–10 day signal cushions but does not avoid a two-week crash; the strategy also gives up upside in sharp recoveries.
- **Moderate statistical confidence.** One market history with roughly four stress episodes; cross-sectional breadth was added, but the number of independent time-series regimes is the binding constraint.

7 Conclusion

The system turns a broken, benchmark-dominated strategy (Sharpe 0.27) into a rigorously validated, risk-managed, alpha-tilted strategy (Sharpe 0.82, Sortino 1.05, max drawdown −8.8%) that demonstrably collapses crisis drawdowns. As much as the numbers, the contribution is the process: leakage control, honest baselines (including discarding an over-fit deep model), robustness and sub-period testing, and explicit limitations.

References

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